

Development of a Deep Learning Model for Analyzing the Optical Properties of Antarctic Ice

with the IceCube Upgrade SKKU Camera System

IceCube Neutrino Observatory

- IceCube is the world's largest neutrino detector, located at the South Pole to detect astrophysical neutrinos. It is composed of 5,160 optical modules (DOMs) installed at depths of approximately 1.5 to 2.5 km within the Antarctic ice, attached to cables known as Strings..

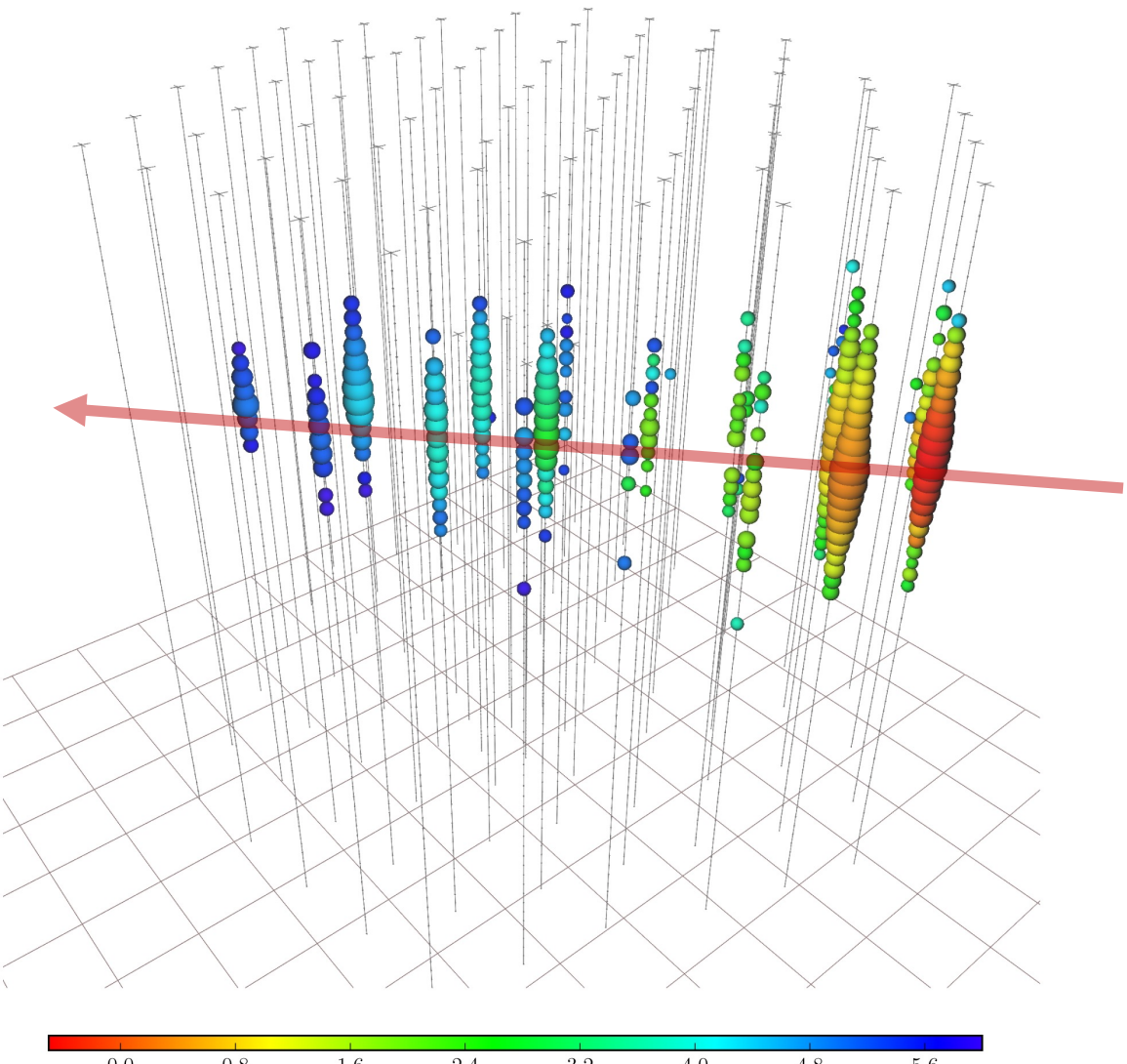
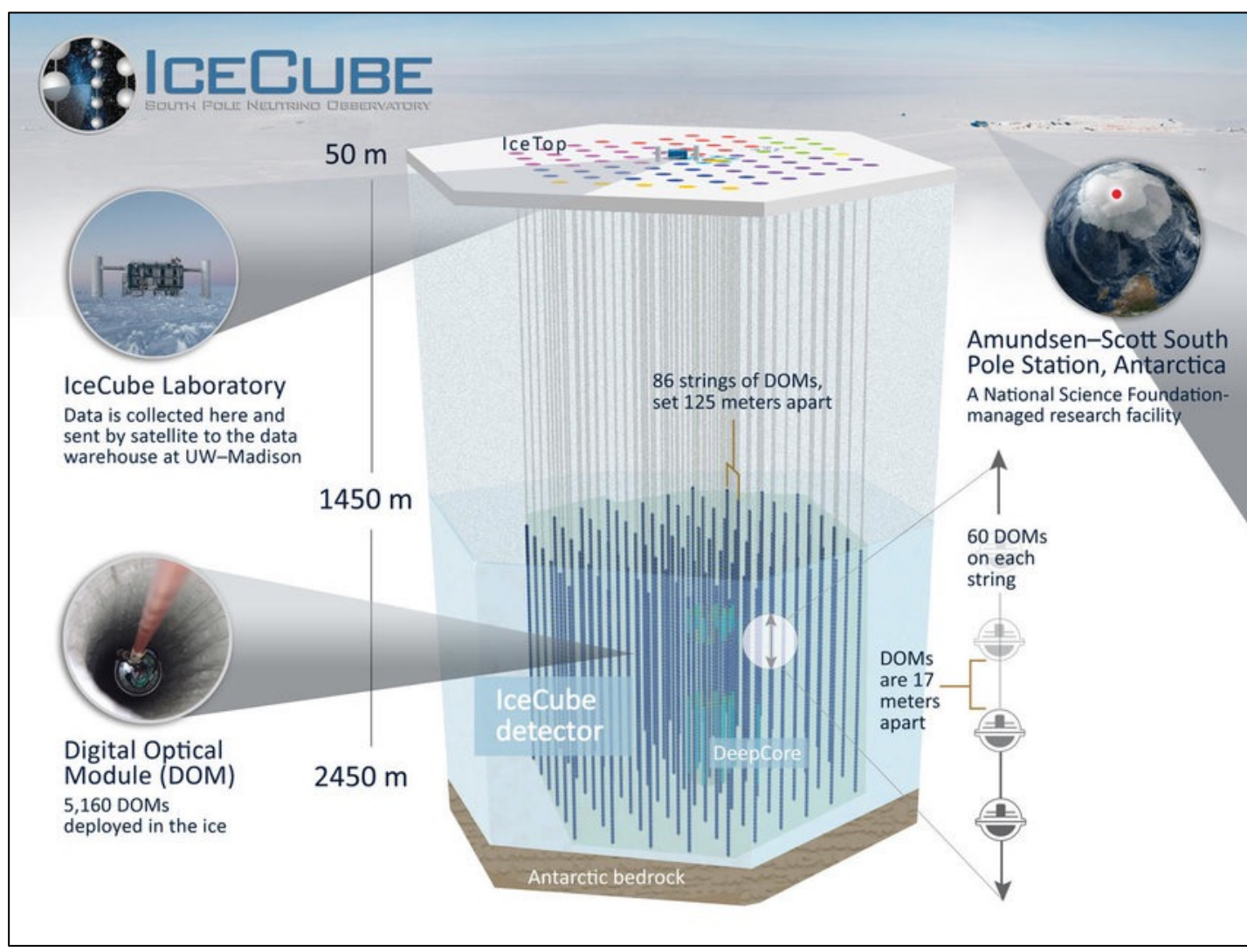
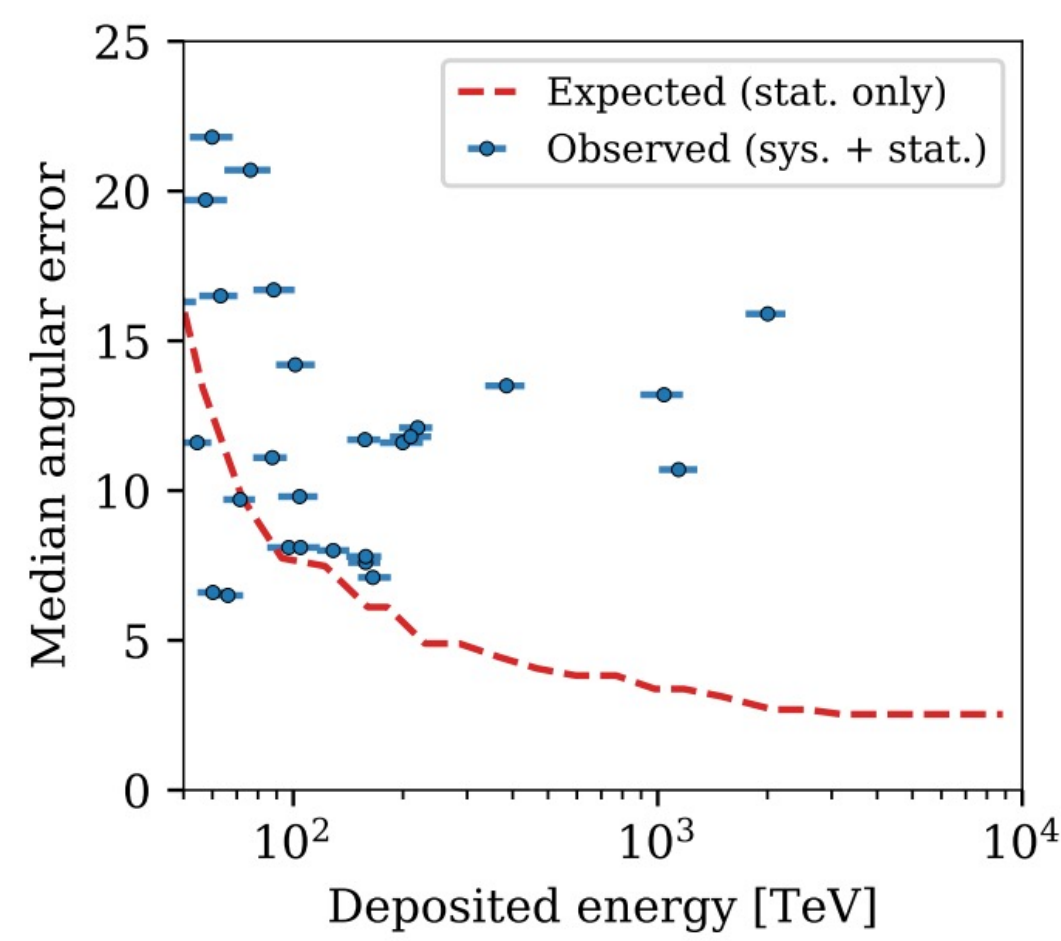


Figure on the left shows the structure of IceCube[1], while the image on the right shows a muon neutrino event observed by IceCube[2].

- IceCube detects neutrinos indirectly through interactions that produce secondary charged particles. These particles emit Cherenkov radiation as they travel through the ice, which is captured by the photomultiplier tubes (PMTs) in the DOMs. By analyzing the signals from all the DOMs, the detector reconstructs the event, allowing estimation of the neutrino's energy, direction, and origin.

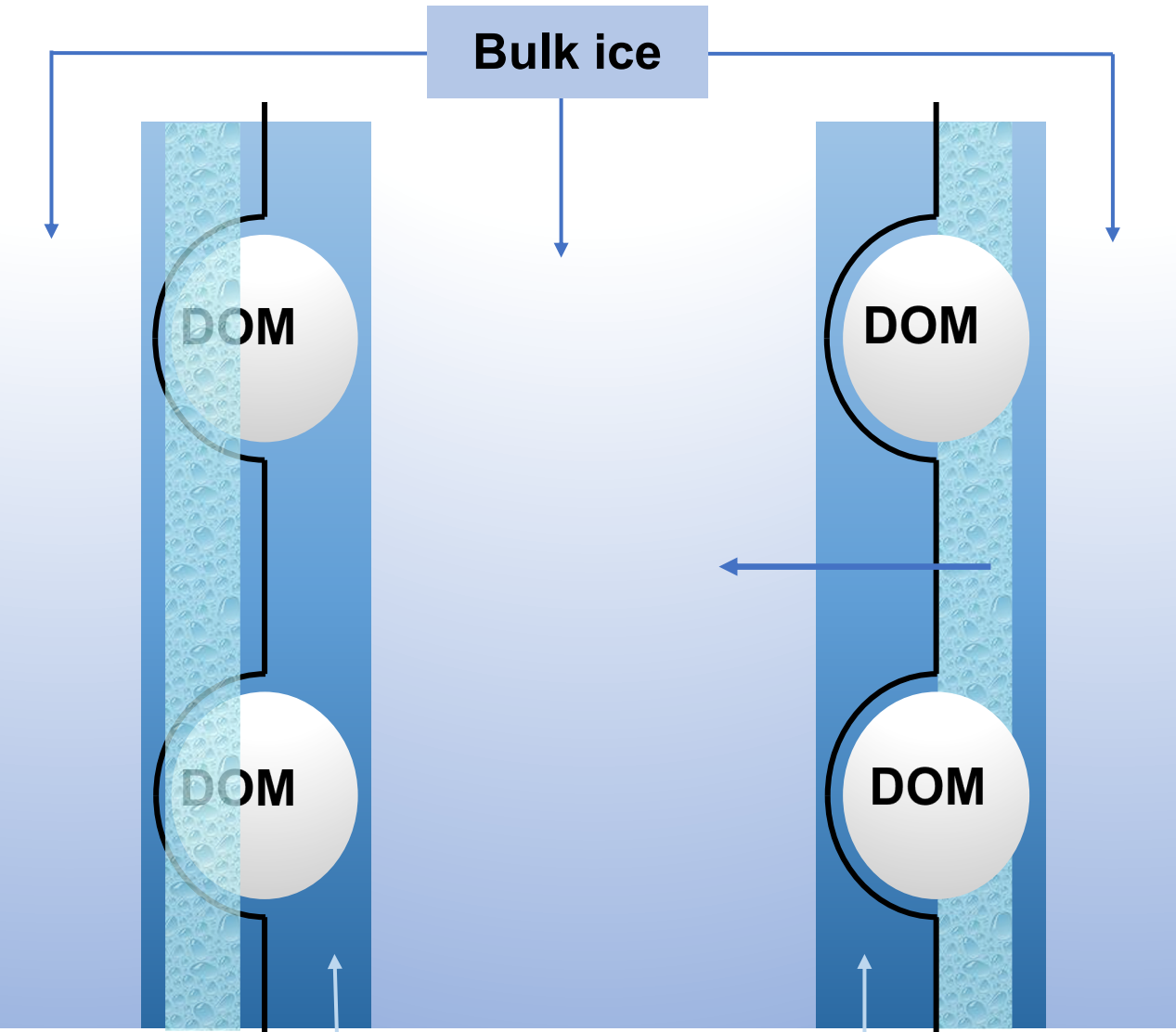
Systematic Uncertainties from Ice



- Since IceCube uses Antarctic ice as the detector medium, the **ice properties are a one of the major source of uncertainties**[3].
- Figure shows observed median angular error of fully contained high energy cascade directional reconstruction as a function of reconstructed deposited energy. The dashed line indicate the reconstruction performance with a perfect knowledge of the optical properties of ice and detector responses. The deviation of data points from the line indicate the presence of incomplete understandings of ice and detector response to bright light[3].

Optical Properties of Ice

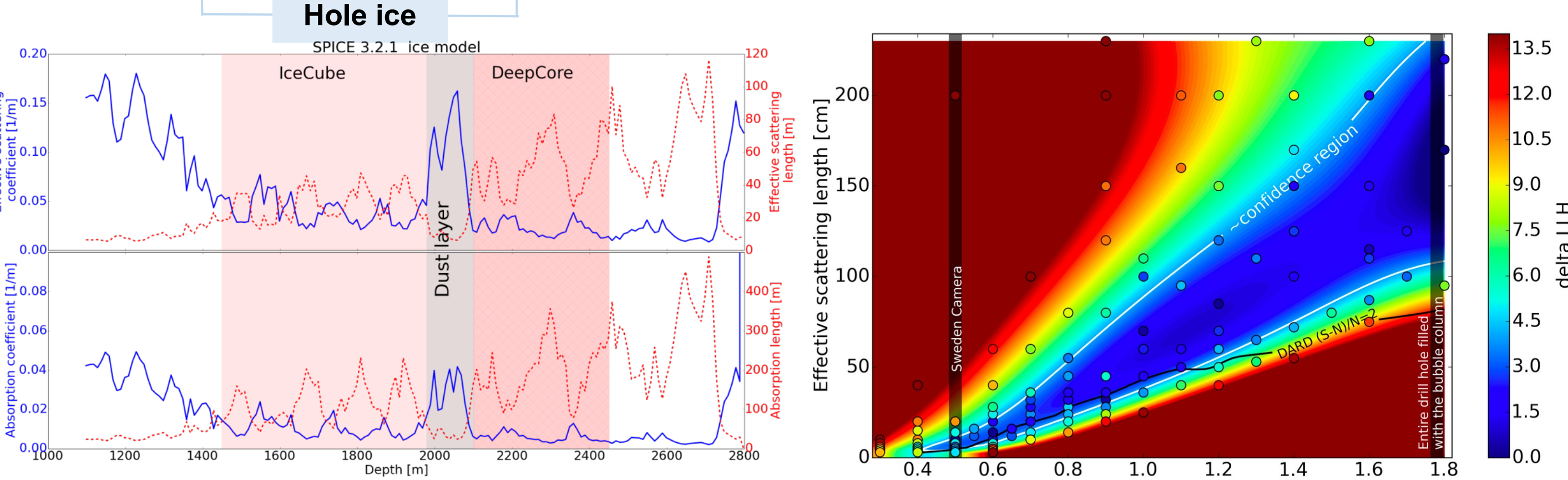
- The Antarctic ice, the detector medium, is divided into two parts: Bulk Ice and Hole Ice.



Bulk Ice: The pure Antarctic glacial ice
Hole Ice: The refrozen ice in which the strings are embedded. It contains residual air bubbles, increasing optical scattering around the DOMs.

Main Bulk Ice Optical Properties	
Scattering Length	Absorption Length

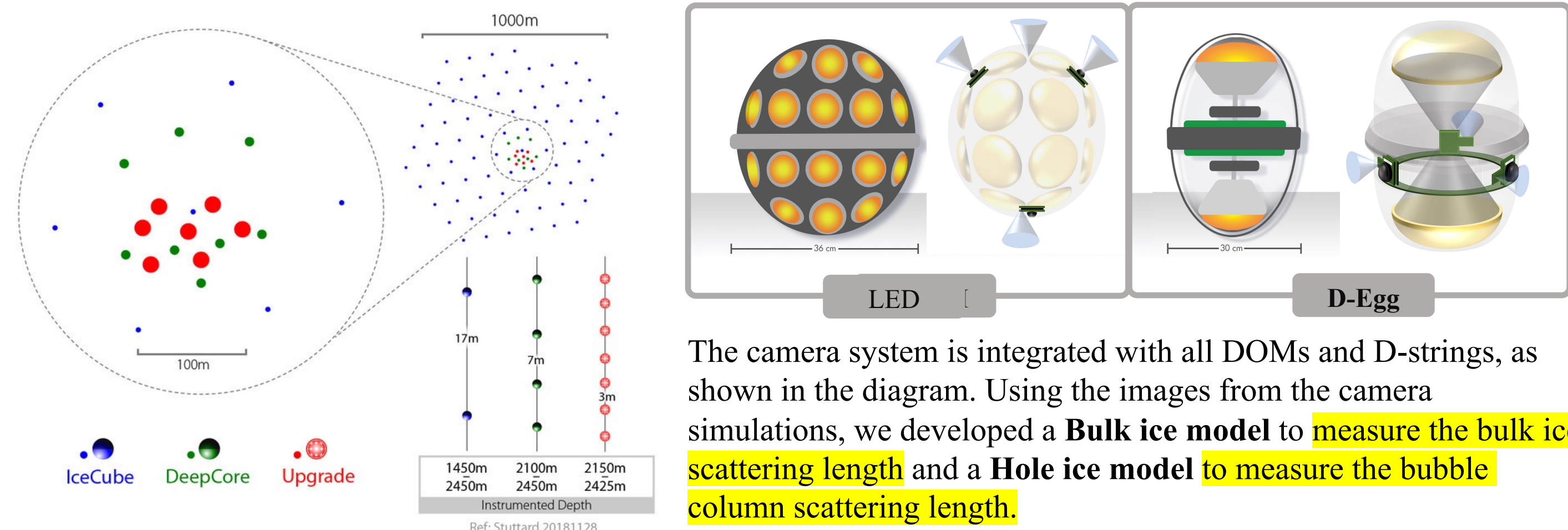
Main Hole Ice Optical Properties		
Scattering Length	Bubble column	
	Size & position	Scattering length



- Figure right shows the distribution of scattering length and absorption length as a function of depth. The model error for the scattering length is known to be approximately 5% ~ 8.5%. Left [5] ikelihood contours based on bubble column size (in DOM units) and scattering strength. Colored circles show delta LLH for tested realizations versus the best fit. Based on these results and the Swedish camera analysis, the radius of the bubble column is estimated to be between 140 to 180 mm, or approximately 0.4-0.55 DOM radii, with the scattering length known to range from about 1 to 10 cm.

IceCube Upgrade and Camera System

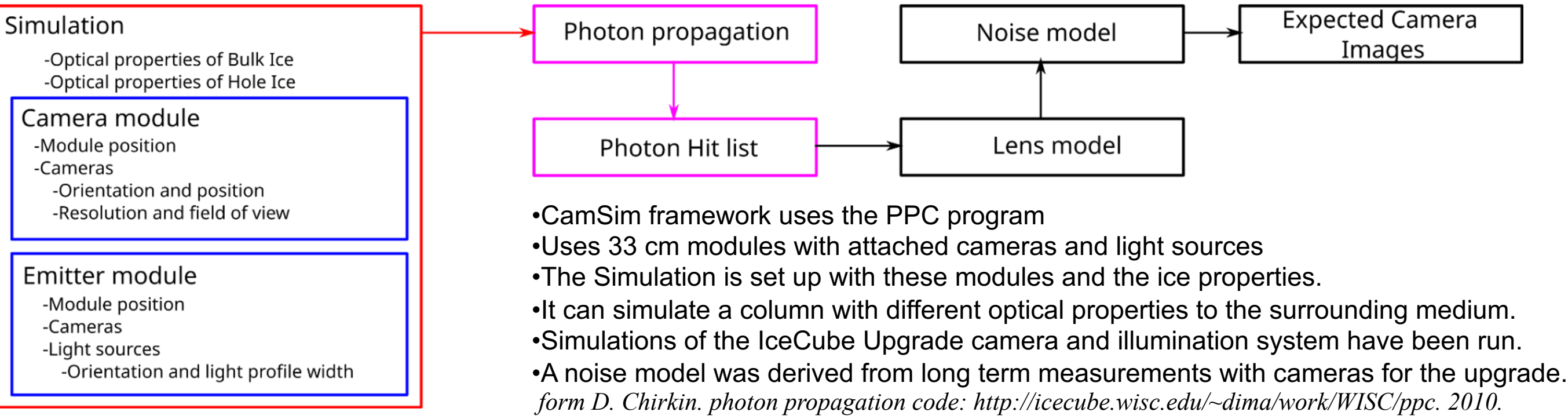
- The IceCube Upgrade project is aiming to deploy 700 densely packed optical modules on 7 new strings in the central region of the detector on 2025~2026.



The camera system is integrated with all DOMs and D-strings, as shown in the diagram. Using the images from the camera simulations, we developed a **Bulk ice model** to **measure the bulk ice scattering length** and a **Hole ice model** to **measure the bubble column scattering length**.

Camera Simulation Framework

- The camera image simulation was created using PPC, which is based on Monte Carlo simulation[6].



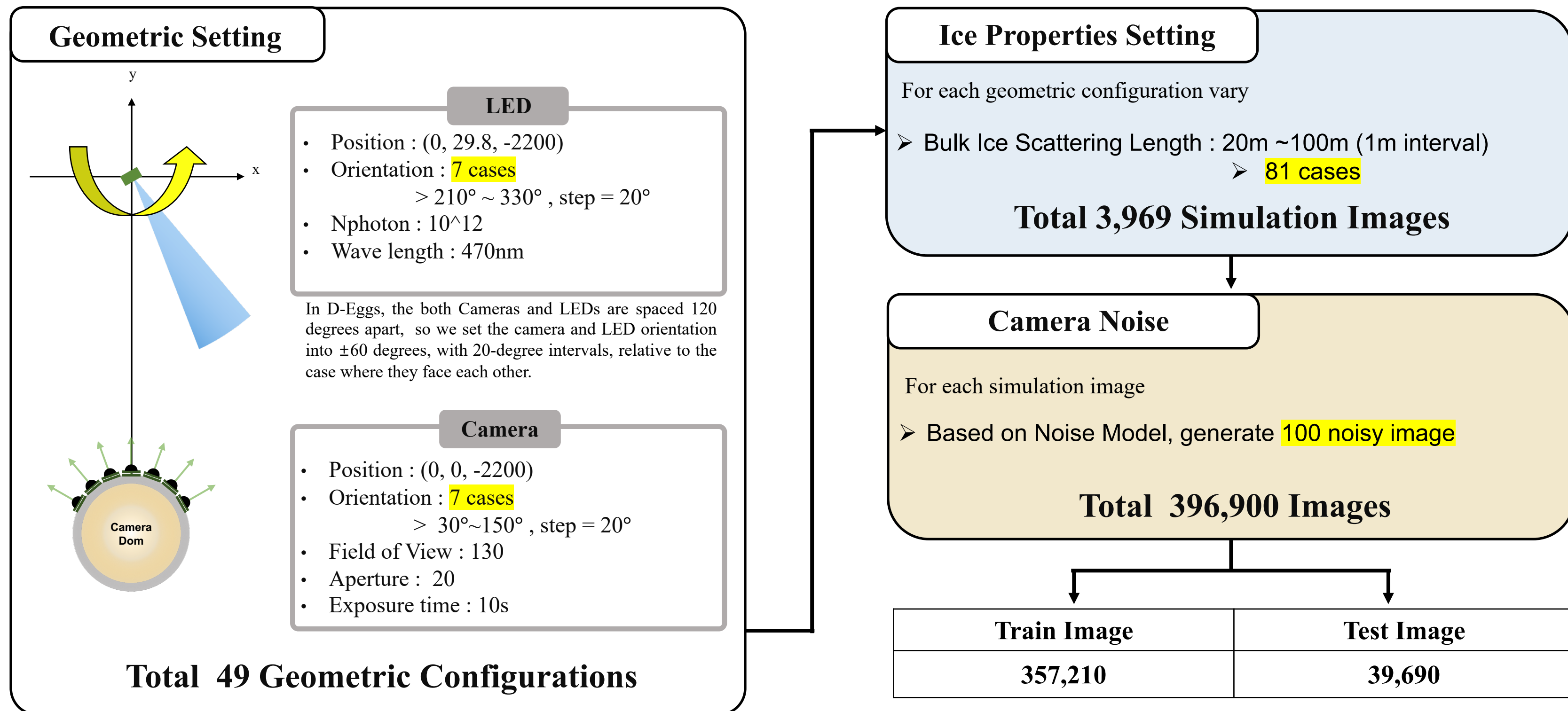
•CamSim framework uses the PPC program
•Uses 33 cm modules with attached cameras and light sources
•The Simulation is set up with these modules and the ice properties.
•It can simulate a column with different optical properties to the surrounding medium.
•Simulations of the IceCube Upgrade camera and illumination system have been run.
•A noise model was derived from long term measurements with cameras for the upgrade.
form D. Chirkin, photon propagation code: <http://icecube.wisc.edu/~dima/work/WISC/ppc>, 2010.

Reference

- [1]IceCube Collaboration. *Improved modeling of in-ice particle showers for IceCube event reconstruction*. Prepared for submission to JINST.
- [2] **IceCube Collaboration**. *IceCube Neutrino Observatory*. Available at: <https://icecube.wisc.edu>
- [3] Henningsen, F., Boehmer, M., Gärtner, A., Geilen, L., Gernhäuser, R., et al. *A self-monitoring precision calibration light source for large-volume neutrino telescopes*.
- [4] IceCube Collaboration. *Improved modeling of in-ice particle showers for IceCube event reconstruction*
- [5] Rongen, Martin. (2019) Calibration of the IceCube Neutrino Observatory (PhD dissertation). RWTH AachenUniversity, Aachen, Germany.
- [6] D. Chirkin. *photon propagation code*: <http://icecube.wisc.edu/~dima/work/WISC/ppc>, 2010
- [7] IceCube Collaboration. *A calibration study of local ice and optical sensor properties in IceCube*. Proceedings of Science, ICRC 2021, The Astroparticle Physics Conference, Berlin, Germany, 12-23 July 2021.

Bulk Ice Model

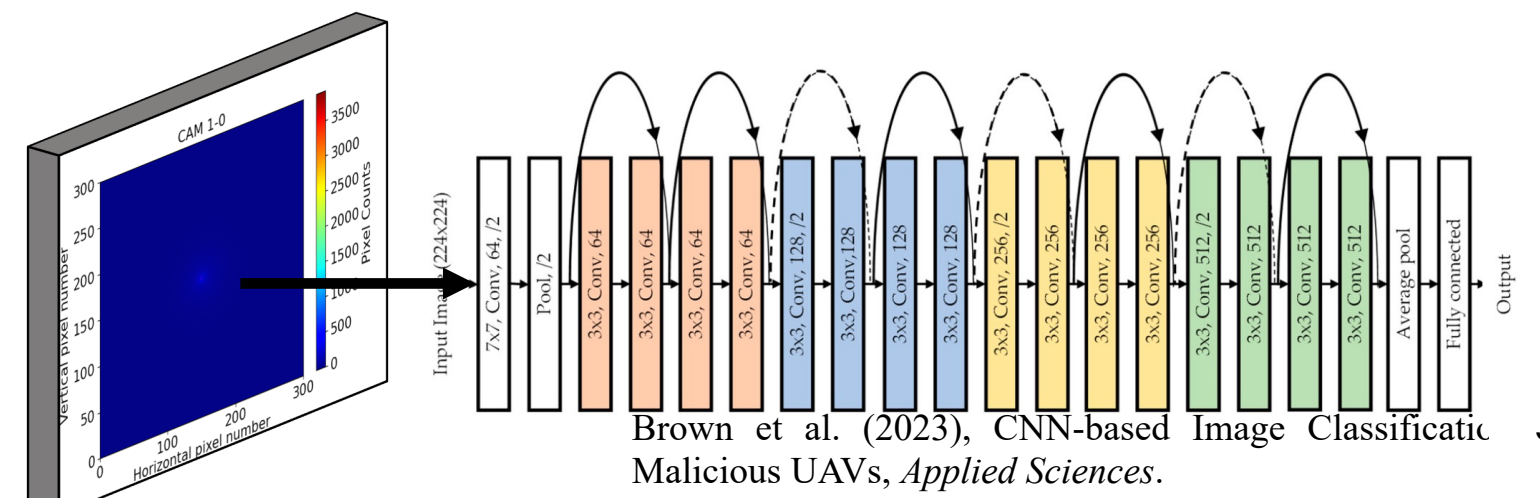
- Objective :** **Measure Bulk Ice Scattering length form the Upgrade Simulation Image.**
- Simulation Setting**



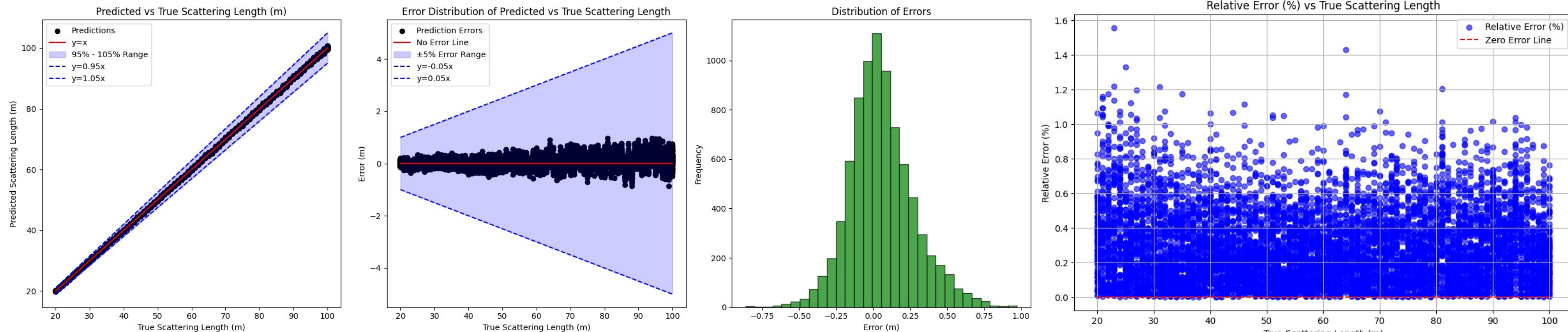
Model Architecture

Base Architecture	AlexNet	ResNet	ViT
N_Layer/Block	3~7	18, 34, 50	2~6

- The results of training models with the architectures listed in the table show that ResNet-18 performed the best.



Performance



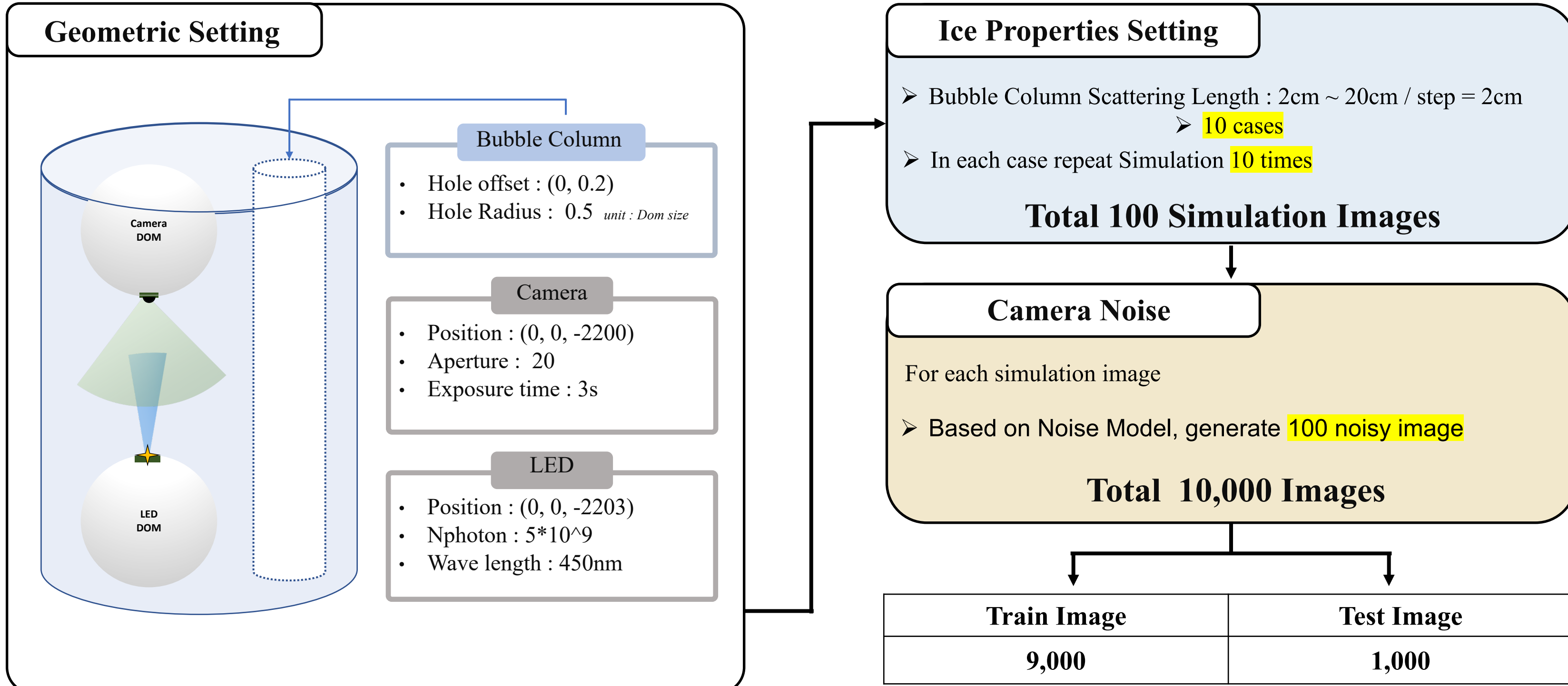
Bulk Ice Scattering Length	20m ~ 40m	40m ~ 60m	60m ~ 80m	80m ~ 100m	Total
Mean Error	0.002m	-0.003m	0.081m	0.139m	0.056m
90% Range of Error	-0.241m~0.303m	-0.400m~0.432m	-0.486m~0.650m	-0.615m~0.854m	-0.512m~0.737m
Mean Relative Error	0.317%	0.261%	0.273%	0.295%	0.286%
90% Range of Relative Error	~1.041%	~0.846%	~0.847%	~0.883%	~0.916%

$$Error(m) = predicted\ scattering\ length(m) - true\ scattering\ length(m)$$

$$Relative\ Error(\%) = \frac{|Error|}{true\ scattering\ length} * 100$$

Hole Ice Model

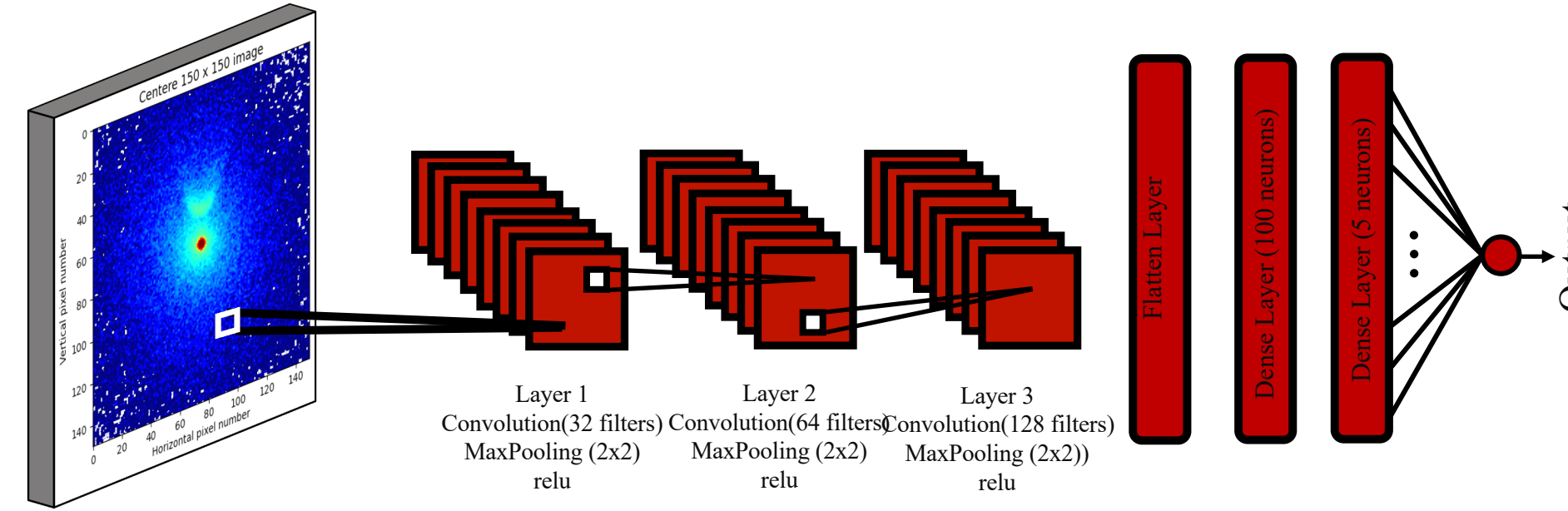
- Objective:** **Measure Bubble Column Scattering length form the Upgrade Simulation Image**
- Simulation Setting**



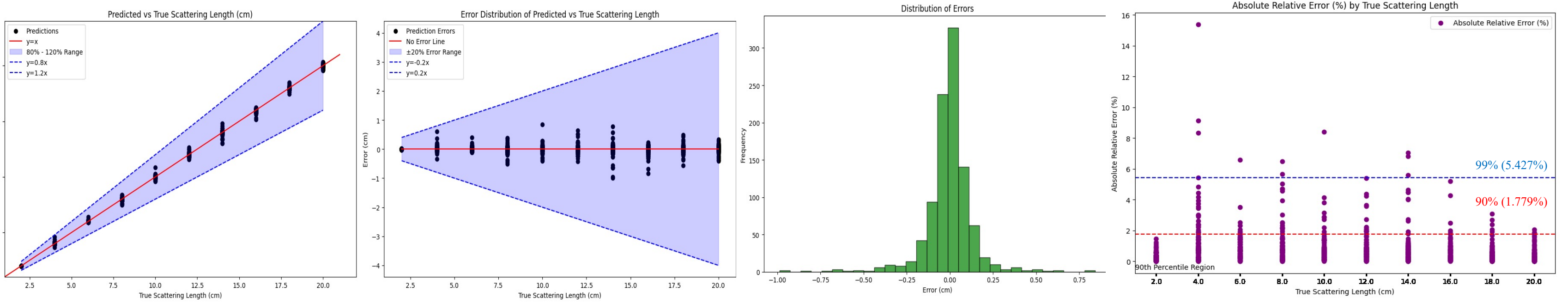
Model Architecture

Base Architecture	N_Layer	Mean Error
AlexNet	3	0.819
AlexNet	4	0.844
AlexNet	5	0.973

- The results of training models with the architectures listed in the table show that AlexNet with 3 Layers performed the best.



Performance



Bubble Column Scattering Length (cm)	2	4	6	8	10	12	14	16	18	20	Total
Mean of Error (cm)	-0.001	0.014	0.013	-0.018	0.003	0.009	-0.020	-0.039	-0.008	0.001	-0.004
Mean of Relative Error (%)	0.342	1.496	0.952	1.119	0.819	0.926	0.968	0.700	0.450	0.448	0.819
90% Range of Error (cm)	±0.013	±0.130	±0.101	±0.187	±0.183	±0.236	±0.263	±0.233	±0.169	±0.217	±0.178
90% Range of Relative Error (%)	1.226	3.268	1.683	2.348	1.835	1.972	1.882	1.458	0.941	0.1085	1.779

$$Error(m) = predicted\ scattering\ length(m) - true\ scattering\ length(m)$$

$$Relative\ Error(\%) = \frac{|Error|}{true\ scattering\ length} * 100$$

Conclusion

Bulk Ice Model	Mean	Max	90% Range
Error	0.056m	0.9739m	-0.512m~0.737m
Relative Error	0.286%	1.556%	~0.916%

Hole Ice Model	Mean	Max	90% Range
Error	-0.004cm	0.841cm	±0.178cm
Relative Error	0.819%	15.371%	~1.779%

- Both models were found to be well-trained without bias.
- For the bulk ice model, even the maximum error is significantly smaller than the previously reported maximum error of 5%, with a value of only 1.5%.
- In the case of the hole ice model, it demonstrates a much lower error compared to the currently proposed likelihood-based methods using other devices[7], which exhibit an error rate of around 20%.